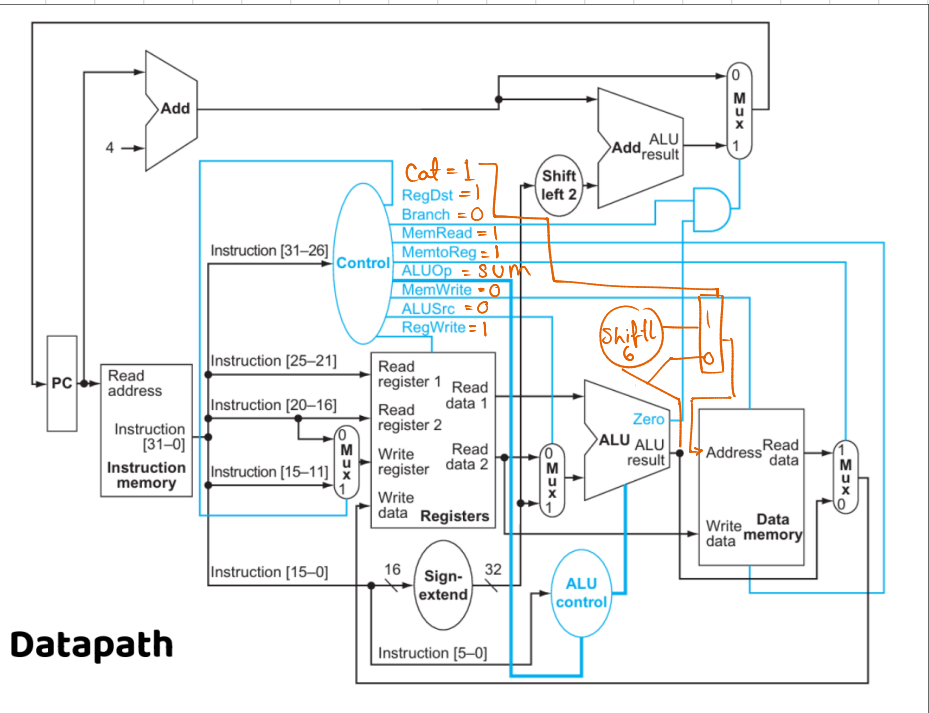


1. (a) You have to modify the block diagram for the single-cycle data path and write the control unit values for this instruction: 'cat \$s0, \$s1, \$s2'. The job of the given instruction is to read the value of the source registers (rs, rt) and add them. After that, the sum will be multiplied by 64 to get an address. We need to read the value of that address and write it to the destination register rd. The machine code format for the given instruction is given below, where rs, rt, and rd will contain the index numbers of \$s1, \$s2, and \$s0 registers, respectively.

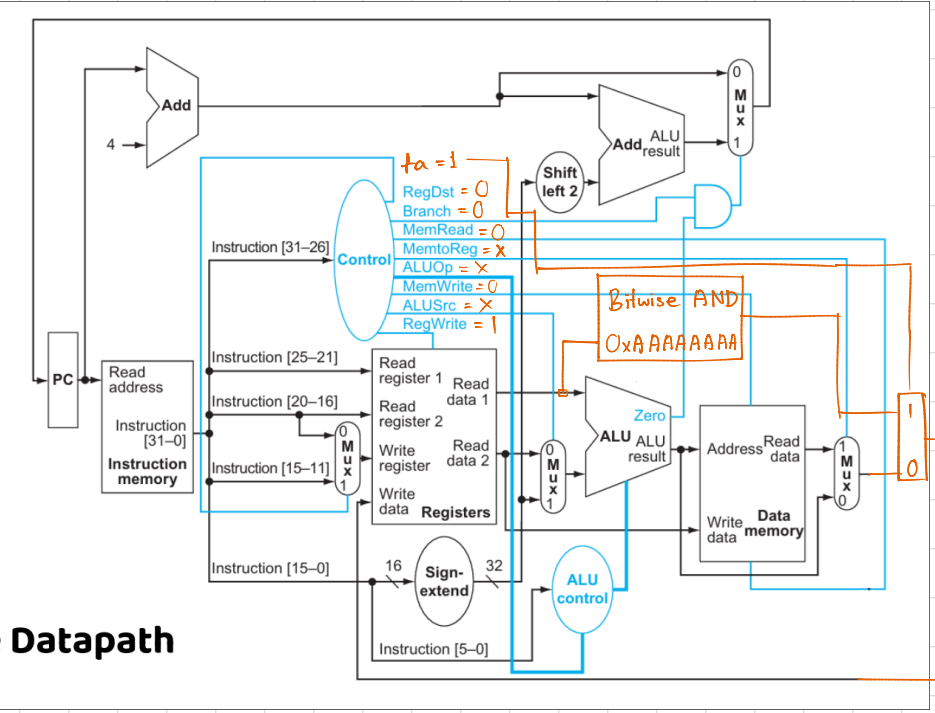
op	rs	rt	rd	shamt	funct
6 bits	5 bits	5 bits	5 bits	5 bits	6 bits



Datapath

(b) Modify the block diagram for a single-cycle data path so that it can execute the following instruction 'fa \$s1, \$s0'. The instruction reads the value of the source register rs and resets(makes 0) all the odd bits of the value. Then the result is written into the destination register rd. Write down the control unit values as well. The machine code format for the given instruction is given below, where rs and rd will contain the index numbers of \$s0 and \$s1 registers, respectively.

op	rs	rd	blank(not used)
6 bits	5 bits	5 bits	16 bits



Datapath

(a) Find out the total time for the given code below in a single cycle method. (Assume \$s0 = 8 and 0(\$s0) holds the value 0)

Listing 1: Code for 2(a)

```
1 lw $t0, 0($s0)
2 add $s3, $zero, $t0
3 addi $s1, $zero, 20
4 addi $s2, $zero, 1
5 LOOP:
6     slt $t1, $s2, $s1
7     beq $t1, $zero, END
8     sub $t2, $s2, $s1
9     add $s3, $s3, $t2
10    sll $s2, $s2, 1
11    srl $s1, $s1, 1
12    j LOOP
13 END:
```

Handwritten notes:
t0=0 s3=0 s1=20 s2=1 s0=8
while s2 < s1
s2 *=2 s2 = 1 -> 2 -> 4 -> 8 break
s1 /=2 s1 = 20 -> 10 -> 5 -> 2 break
LOOP runs 3 times

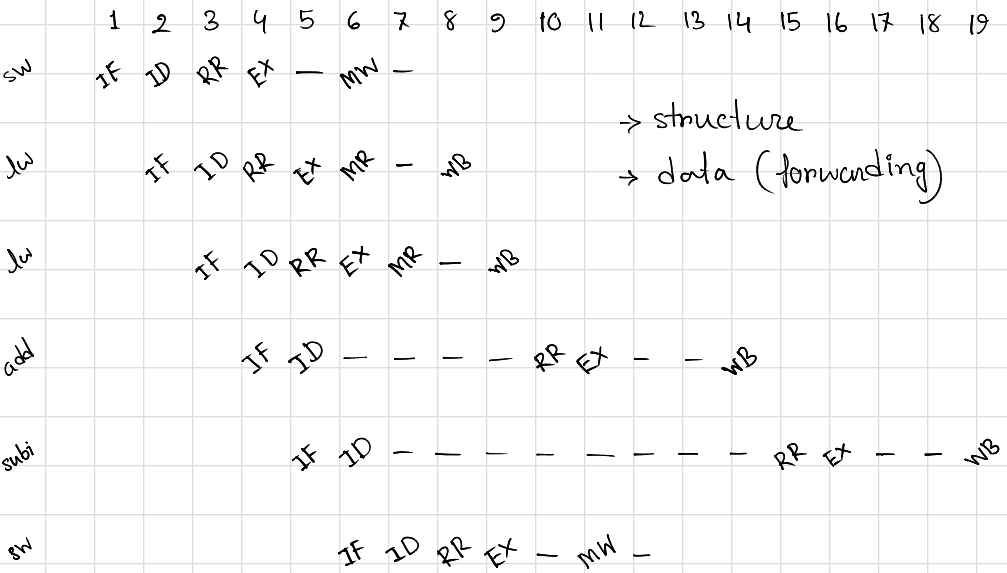
Number of instructions = 4 + 3*7 + 2 = 27

Time per cycle = 1080 ns

Total time = 27 * 1080 = 29160 ns

```
1 sw $t0, 0($s0)
2 lw $t1, 4($s0)
3 lw $t2, 8($s0)
4 add $t2, $t1, $t2
5 subi $t2, $t2, 10
6 sw $t4, 8($s0)
```

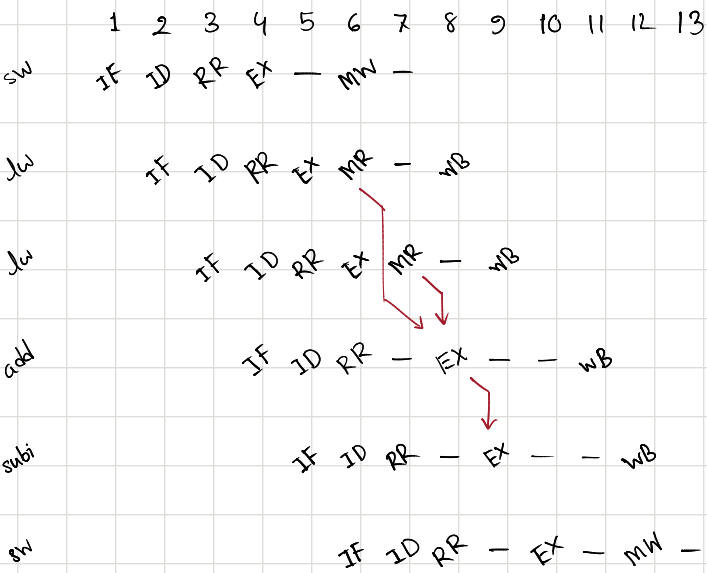
(b) Implement the basic pipeline method and find out the total time using a timing diagram.



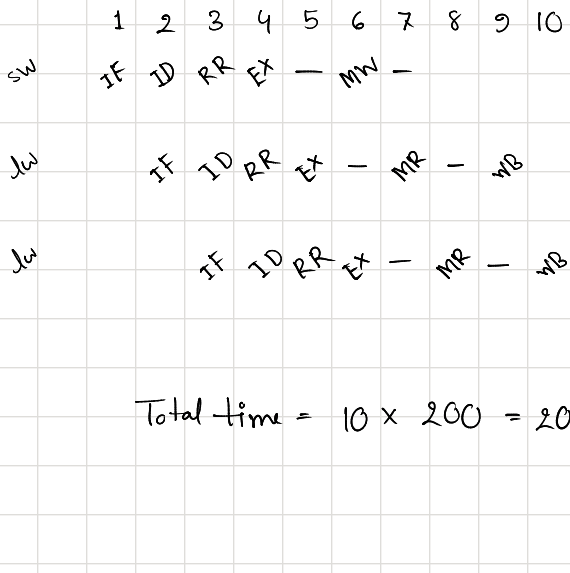
Total time = 19 * 200 = 3800 ns

```
1 sw $t0, 0($s0)
2 lw $t1, 4($s0)
3 lw $t2, 8($s0)
4 add $t2, $t1, $t2
5 subi $t2, $t2, 10
6 sw $t4, 8($s0)
```

(b) Implement the basic pipeline method and find out the total time using a timing diagram.



Total time = 13 * 200 = 2600



Total time = 10 * 200 = 2000 ns

3. A 16-bit 2GHz processor has a cache memory of 128 bytes, which is byte-addressable, and a line size of 2 words. Each word contains 2 bytes of data. Note that cache memory contains two flag bits: a valid bit and a dirty bit. Based on the information, answer the following questions:

(a) **Find** out index bits and tag bits.

[2]

(b) **Calculate** the actual size of the cache (in Bytes).

[3]

a) Address 16 bit ; # Cache Lines = $\frac{128}{4} = 32 = 2^5$

9 bit	5 bit	1 bit	1 bit
Tag	Index	valid	dirty

b) Cache Line:

Tag	Valid	Dirty	Data
9b	1b	1b	4B

Actual size = $(9 + 1 + 1 + 4 \times 8) \times 32 = 1376 \text{ b} = 172 \text{ B}$